

Probability Homework 4

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3.2-17

The pdf of chi-square distribution is:

$$f(x) = \frac{1}{\Gamma(r/2)2^{r/2}} x^{r/2-1} e^{-x/2}$$

In this problem $r = 4$, so $f(x) = \frac{1}{4}xe^{-x/2}$. Let X be the r.v. of $\chi^2(4)$.

$$\begin{aligned} P(X > 7.779) &= 1 - P(X \leq 7.779) \\ &= 1 - \int_0^{7.779} \frac{1}{4}xe^{-x/2}dx \\ &\approx 1 - 0.9 = 0.1 \end{aligned}$$

Let Y be the number of observations (out of 15) that exceeds 7.779, since the observations are taken independently, it can be seen as a binomial distribution with $p = 0.1, n = 15$.

$$\begin{aligned} P(Y \leq 3) &= \sum_{k=0}^3 \binom{15}{k} (0.1)^k (0.9)^{15-k} \\ &= (0.9)^{15} + 15 \cdot (0.1)(0.9)^{14} + 105 \cdot (0.1)^2(0.9)^{13} + 455 \cdot (0.1)^3(0.9)^{12} \\ &= 0.9444 \end{aligned}$$

Hence, the probability that at most 3 of the 15 observations exceed 7.779 is 0.9444.

3.2-18

Let the r.v. $Y = X - 80$, so $Y > 0$ with pdf:

$$f(y) = \frac{y}{50^2} e^{-y/50}, \quad y > 0$$

This is Gamma distribution with shape $\alpha = 2$ and scale $\theta = 50$.

(a)

$$\begin{aligned}\mu_Y = E[Y] &= \alpha\theta = 2 \cdot 50 = 100 \\ \mu_X = E[X] &= E[Y + 80] = E[Y] + 80 = 180\end{aligned}$$

Shifting doesn't affect the variance:

$$\sigma_X^2 = \sigma_Y^2 = \alpha\theta^2 = 5000$$

Hence, the mean is 180 and the variance is 5000.

(b)

We can find the mode by setting $f'(y) = 0$:

$$\begin{aligned}f'(y) &= \frac{1}{2500}(e^{-y/50} - \frac{1}{50}ye^{-y/50}) \\ &= \frac{1}{2500}e^{-y/50}(1 - \frac{1}{50}y) = 0 \\ \Rightarrow y &= 50\end{aligned}$$

Hence, the mode of X is $50 + 80 = 130$.

(c)

This is equivalent to finding $P(Y < 120) = 1 - P(Y \geq 120)$:

$$\begin{aligned}P(Y < 120) &= 1 - \int_{120}^{\infty} \frac{y}{50^2} e^{-y/50} dy \\ &= 1 - \frac{1}{2500}(-50ye^{-y/50} \Big|_{120}^{\infty} + 50 \int_{120}^{\infty} e^{-y/50} dy) \\ &= 0.6915\end{aligned}$$

The percentage of having a serum cholesterol level less than 200 is 0.6915.

3.3-11

We are given that $\mu = 21.37, \sigma^2 = 0.16, \sigma = 0.4$.

(a)

Standardize using $Z = \frac{X-\mu}{\sigma}$.

$$P(X > 22.07) = P(Z > 1.75) = 1 - \Phi(1.75) = 1 - 0.9599 = 0.0401$$

where the value of $\Phi(1.75)$ is obtained by looking up the table.

We obtained that $P(X > 22.07) = 0.0401$

(b)

$$\begin{aligned} P(X < 20.857) &= P(Z < -1.28) \\ &= P(Z \leq -1.28) + P(Z = 1.28) \\ &= [1 - \Phi(1.28)] + \int_{1.28}^{1.28} f(x)dx \\ &= 1 - \Phi(1.28) = 1 - 0.8997 = 0.1003 \end{aligned}$$

Since each selection is independent, it can be seen as a binomial distribution with $p = 0.1$ and $n = 15$. (Y is the r.v.)

$$\begin{aligned} P(Y \leq 2) &= P(Y = 0) + P(Y = 1) + P(Y = 2) \\ &= \binom{15}{0}(0.9)^{15} + \binom{15}{1}(0.1)(0.9)^{14} + \binom{15}{2}(0.1)^2(0.9)^{13} \\ &= 0.8159 \end{aligned}$$

So we have that $P(Y \leq 2) = 0.8159$.

4.1-9

(a)

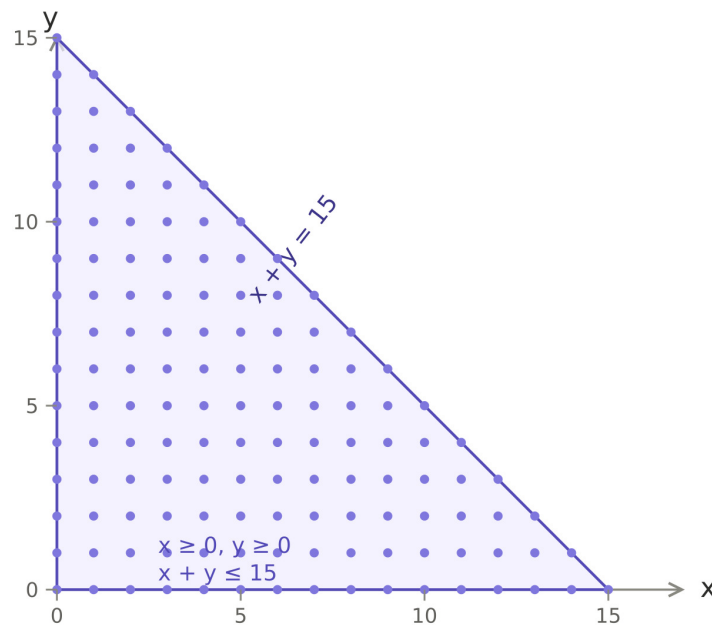
For 15 independent items, $(X, Y, 15 - X - Y)$ follows a multinomial distribution. So the joint pmf is:

$$f(x, y) = P(X = x, Y = y) = \frac{15!}{x!y!(15-x-y)!} \left(\frac{6}{10}\right)^x \left(\frac{3}{10}\right)^y \left(\frac{1}{10}\right)^{15-x-y}$$

And it's valid when $x \geq 0, y \geq 0, x + y \leq 15$. (Otherwise, $f(x, y) = 0$)

(b)

By the constraints I previously mentioned, we can draw the following graph. (The colored region is the set of integers (x, y) for which $f(x, y) > 0$)



Independence would require a rectangular support, so X and y are NOT independent. This is trivial because if X and Y are independent, we have:

$$f(x, y) = f_X(x)f_Y(y)$$

So $f_X(x) > 0, f_Y(y) > 0$ implies $f(x, y) > 0$, this forces the set of points to be the Cartesian product of X and Y , the region must be rectangular. (However, in this problem, $(x, y) = (15, 15)$, $f(x, y) \neq f_X(x)f_Y(y)$ obviously)

(c)

$$\begin{aligned} P(X = 10, Y = 4) &= f(10, 4) \\ &= \frac{15!}{10!4!1!} \cdot (0.6)^{10} \cdot (0.3)^4 \cdot (0.1)^1 \\ &= 0.0735 \end{aligned}$$

(d)

We should sum over all possible y , but we each item is "good" with probability $6/10$, regardless of other categories. So we can view it as the binomial distribution of $n = 15$ and $p = \frac{6}{10}$ (where p is the probability of having "good" item)

Thus, the marginal pmf of X is:

$$f_X(x) = P(X = x) = \binom{15}{x} (0.6)^x (0.4)^{15-x}$$

(e)

$$\begin{aligned} P(X \leq 11) &= 1 - P(X > 11) \\ &= 1 - (P(X = 12) + P(X = 13) + P(X = 14) + P(X = 15)) \\ &= 1 - \left(\binom{15}{12} (0.6)^{12} (0.4)^3 + \binom{15}{13} (0.6)^{13} (0.4)^2 + \binom{15}{14} (0.6)^{14} (0.4)^1 + \binom{15}{15} (0.6)^{15} \right) \\ &= 0.9095 \end{aligned}$$

4.2-3

Theorem 0.1. *The expected values of two random variables is multiplicative if **they are independent**, i.e., $E[XY] = E[X]E[Y]$*

Proof. If X and Y are independent (and $\mathcal{X}$ and $\mathcal{Y}$ are the sample spaces for X and Y respectively.), it holds that

$$p(x, y) = p(x)p(y) \quad \text{for all } x \in \mathcal{X}, y \in \mathcal{Y}. \quad (1)$$

Applying this to the expected value for discrete random variables, we have

$$\begin{aligned} E(XY) &= \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} (x \cdot y) \cdot f_{X,Y}(x, y) \\ &= \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} (x \cdot y) \cdot (f_X(x) \cdot f_Y(y)) \\ &= \sum_{x \in \mathcal{X}} x \cdot f_X(x) \sum_{y \in \mathcal{Y}} y \cdot f_Y(y) \\ &= \sum_{x \in \mathcal{X}} x \cdot f_X(x) \cdot E(Y) \\ &= E(X)E(Y). \end{aligned}$$

And applying it to the expected value for continuous random variables, we have

$$\begin{aligned} E(XY) &= \int_{\mathcal{X}} \int_{\mathcal{Y}} (x \cdot y) \cdot f_{X,Y}(x, y) dy dx \\ &= \int_{\mathcal{X}} \int_{\mathcal{Y}} (x \cdot y) \cdot (f_X(x) \cdot f_Y(y)) dy dx \\ &= \int_{\mathcal{X}} x \cdot f_X(x) \int_{\mathcal{Y}} y \cdot f_Y(y) dy dx \\ &= \int_{\mathcal{X}} x \cdot f_X(x) \cdot E(Y) dx \\ &= E(X)E(Y). \end{aligned}$$

□

Theorem 0.2. *Suppose X and Y are two random variables, then:*

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$

Proof. We can follow the definition of variance and covariance, and writes:

$$\begin{aligned} \text{Var}(X + Y) &= E[(X + Y - \mu_{X+Y})^2] \\ &= E[(X - \mu_X) + (Y - \mu_Y)]^2 \\ &= E[(X - \mu_X)^2] + E[(Y - \mu_Y)^2] + 2E[(X - \mu_X)(Y - \mu_Y)] \\ &= \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y) \end{aligned}$$

The proof is completed here.

□

Theorem 0.3. Suppose X and Y are two independent random variables, then $\text{Cov}(X, Y) = 0$.

Proof. By the formula of covariance and theorem 0.1, we have:

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = E[X] \cdot E[Y] - E[X] \cdot E[Y] = 0$$

□

(a)

Let Z be the outcome of the second roll:

$$\mu_X = E[X] = \mu_Z = E[Z] = \frac{1 + 2 + 3 + 4}{4} = 2.5$$

Since $Y = X + Z$:

$$\mu_Y = E[Y] = E[X + Z] = E[X] + E[Z] = 5$$

Then we can calculate σ_X^2 and σ_Y^2 :

$$\begin{aligned}\sigma_X^2 &= E[X^2] - (E[X])^2 \\ &= \sum_{k=1}^4 k^2/4 - (2.5)^2 \\ &= 1.25 \\ \sigma_Y^2 &= \sigma_X^2 + \sigma_Z^2 + 2\text{Cov}(X, Z) \\ &= \sigma_X^2 + \sigma_Z^2 = 1.25 + 1.25 = 2.5\end{aligned}$$

The last equation comes from theorem 0.2.

Finally, we can calculate the covariance between X and Y using the following steps:

$$\begin{aligned}\text{Cov}(X, Y) &= \text{Cov}(X, X + Z) \\ &= E[X(X + Z)] - E[X]E[X + Z] \\ &= E[X^2] + E[XZ] - (E[X])^2 - E[X]E[Z] \\ &= E[X^2] - (E[X])^2 + E[XZ] - E[X]E[Z] \\ &= \sigma_X^2 + 0 = 1.25 \\ \rho &= \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} \\ &= \frac{1.25}{\sqrt{1.25}\sqrt{2.5}} \\ &= 0.7071\end{aligned}$$

(b)

Suppose the regression line is $y = mx + c$, then slope m and intercept c are:

$$\begin{aligned}m &= \rho \cdot \frac{\sigma_Y}{\sigma_X} = 0.7071 \cdot \frac{\sqrt{2.5}}{\sqrt{1.25}} \approx 1 \\ c &= \mu_Y - \mu_X = 2.5\end{aligned}$$

Hence, the line is $y = x + 2.5$. It's very intuitive because the regression line is exactly taking the first roll (X), then add the average of the second roll (Z with $E[Z] = 2.5$).

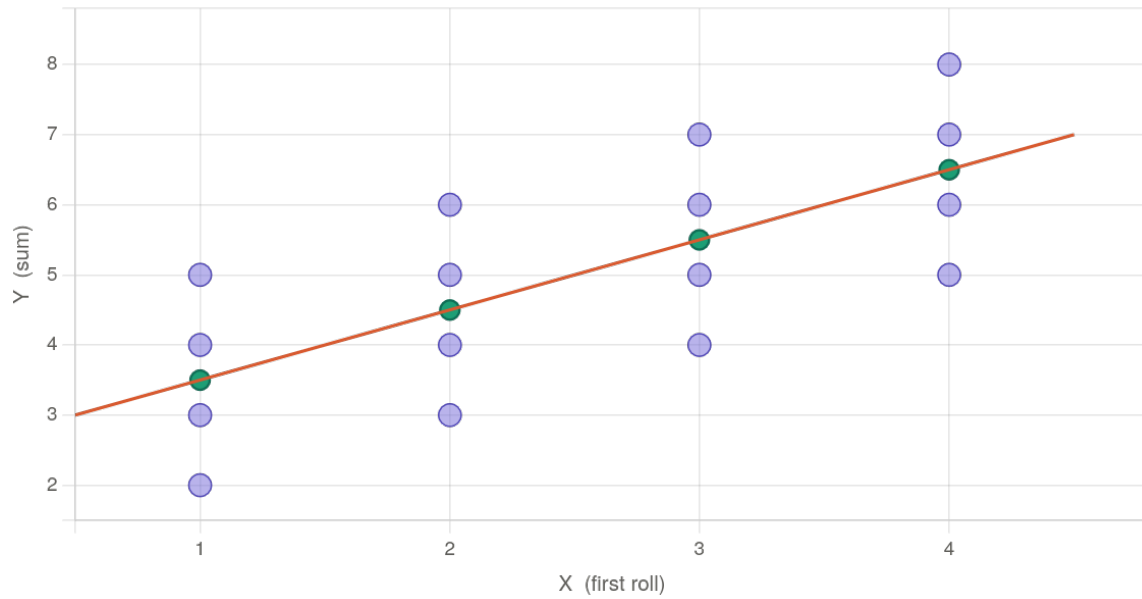


Figure 1: The least square regression line ($y = x + 2.5$)

The purple dots are all the possible outcomes. The green dots show the conditional means $E[Y|X = x] = x + 2.5$ for each x , and the regression line (The red line) passes exactly through all four of them, this is a beautiful confirmation that the line $\hat{y} = x + 2.5$ is correct